

Speak Like A Mathematician:

vector:

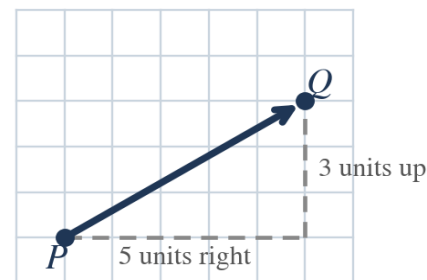
component form:

transformation:

preimage / image:

translation:

rigid motion:



Vectors and Component Form

The _____ point of a vector is where it starts; the _____ point is where it ends.

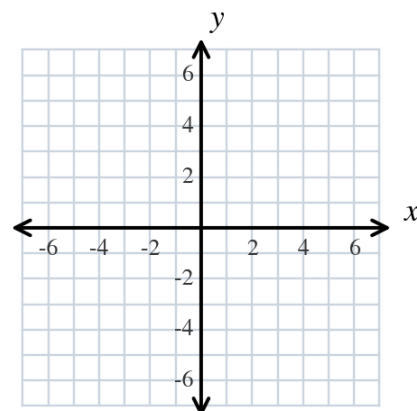
The component form $\langle a, b \rangle$ records the _____ change and the _____ change.

Translation Postulate: a translation is a _____, so the image is congruent to the preimage.

Example 1 - Translate a Figure Along a Vector

Translate $\triangle ABC$ with $A(-4, 2)$, $B(1, 4)$, and $C(3, -1)$ along the vector $\langle 2, -3 \rangle$. Graph the preimage and the image.

- Write the translation as a rule.
- Find the coordinates of the image.
- Is $A'B'C'$ congruent to ABC ? Explain.



Writing and Using a Translation Rule

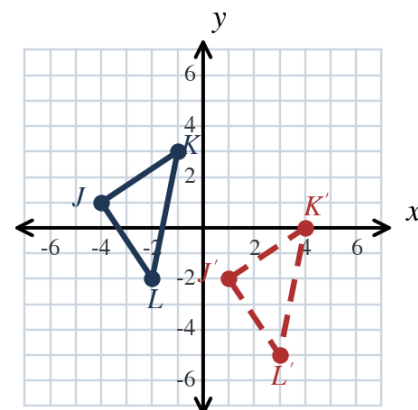
A translation along $\langle a, b \rangle$ is written $(x, y) \rightarrow (\underline{\hspace{2cm}}, \underline{\hspace{2cm}})$.

To go from a point to its image, the components. To go BACKWARD from an image to its preimage, them.

Example 2 - Find the Component Form of a Vector

Find the component form of the vector that translates $M(3, -1)$ to M' .

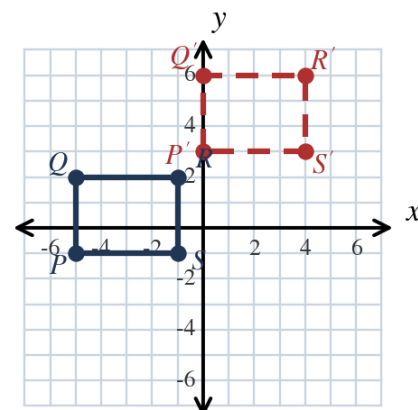
- a) $M'(-2, 4)$
- b) $M'(7, -6)$
- c) $M'(0, 2)$



Example 3 - Write a Rule for a Translation

Write a rule for the translation of the blue preimage to the red image.

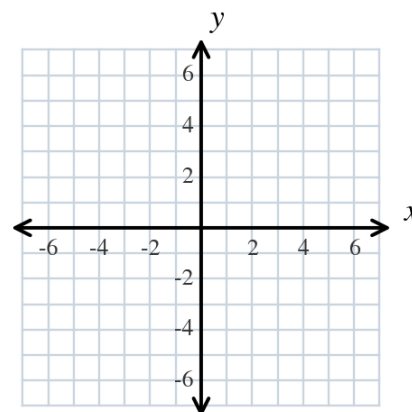
- a) Use $\triangle JKL$ and $\triangle J'K'L'$.
- b) Use rectangle PQRS and $P'Q'R'S'$.
- c) Describe the vector for part (a) in words.



Example 4 - Using a Translation Rule

Use the translation $(x, y) \rightarrow (x - 4, y + 5)$.

- a) What is the image of $A(2, -3)$?
- b) What is the image of $B(-6, 1)$?
- c) What is the preimage of $C'(3, 0)$?
- d) What is the preimage of $D'(-1, -4)$?



Example 5 - A Composition of Two Translations

Graph $\triangle RST$ with $R(-3, 2)$, $S(0, -1)$, and $T(-4, -3)$ and its image after the composition.

Translation: $(x, y) \rightarrow (x + 6, y - 2)$

Translation: $(x, y) \rightarrow (x - 1, y + 5)$

- a) Find the coordinates after the first translation.
- b) Find the coordinates after the second translation.
- c) Write ONE rule for the whole composition.

Example 6 - A Real-Life Application

A game piece sits at $(2, 5)$. You move it along the vector $\langle -3, -4 \rangle$.

- a) Find the new position of the piece.
- b) How far did the piece move? Use the Distance Formula.
- c) A second piece at $(-6, 0)$ is moved along the same vector. How far does it move? Explain.